Topics of today's class

Last class:

- Chapter 7
  - Economic growth facts
  - Malthusian model of economic growth
  - Solow growth model

Today's class:

- Chapter 8
  - Income disparity among countries
  - Endogenous growth model

Chapter 8

Income Disparity Among Countries and Endogenous Growth

Chapter 8 Topics

- Convergence
- Endogenous growth: a model of human capital accumulation

Convergence in the Solow Growth Model

If two countries are initially rich and poor, but identical in all other respects, they will converge in the long run to the same level and rate of growth of per-capita income.

Figure 8.1
Rich and Poor Countries and the Steady State

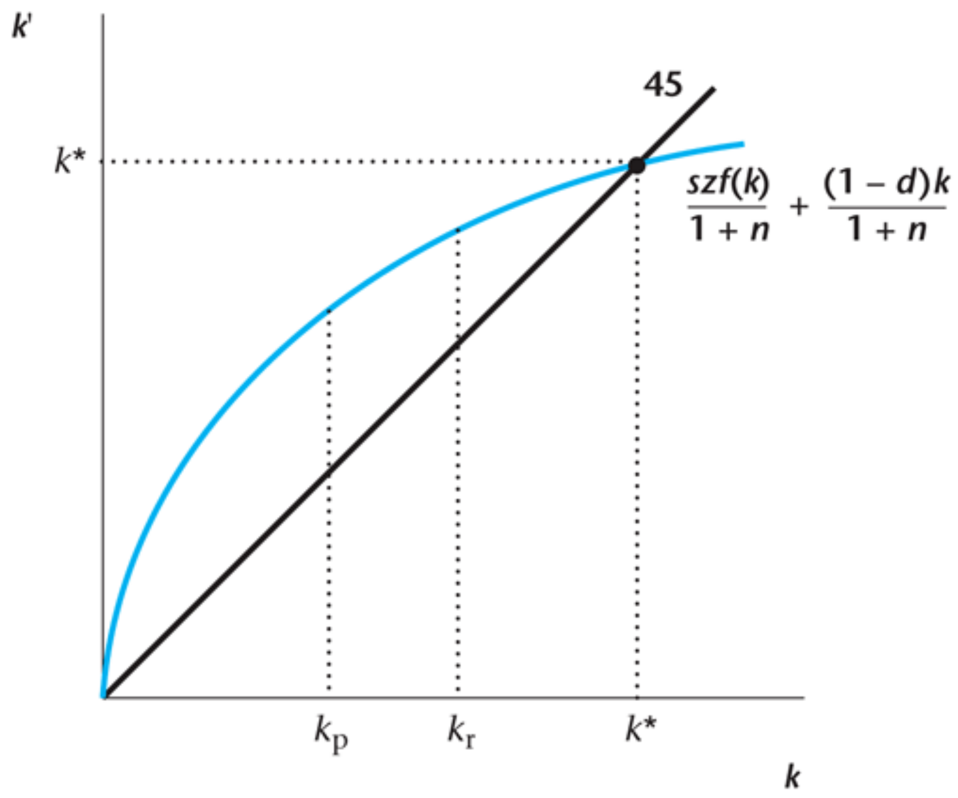


Figure 8.2
Convergence in Income per Worker Across Countries in the Solow Growth Model

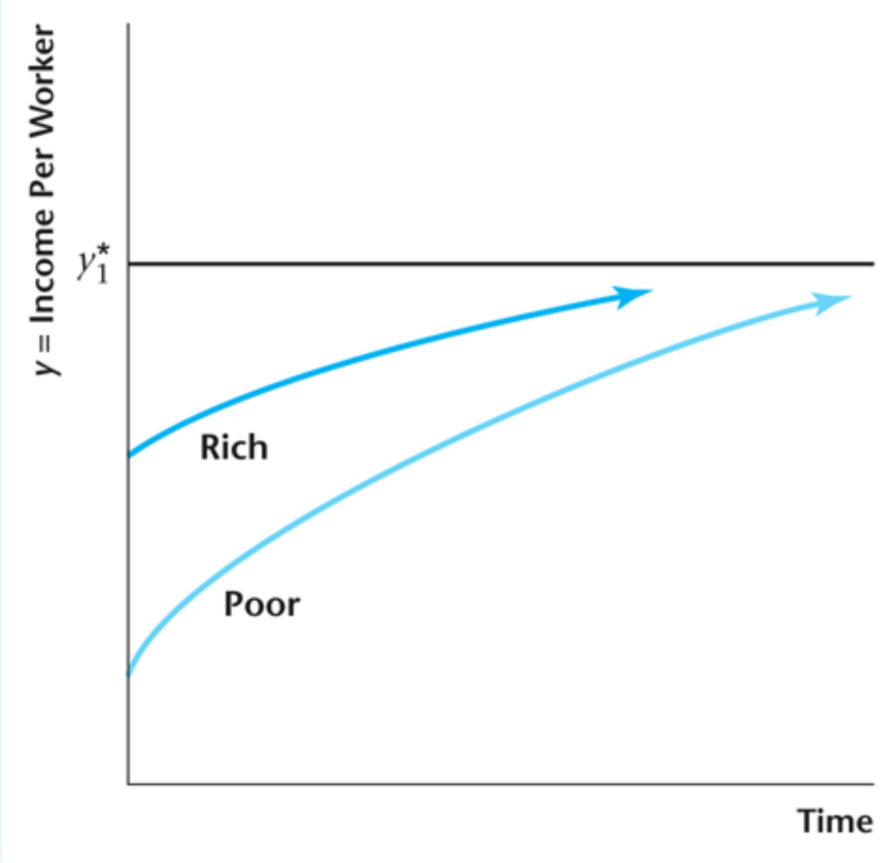
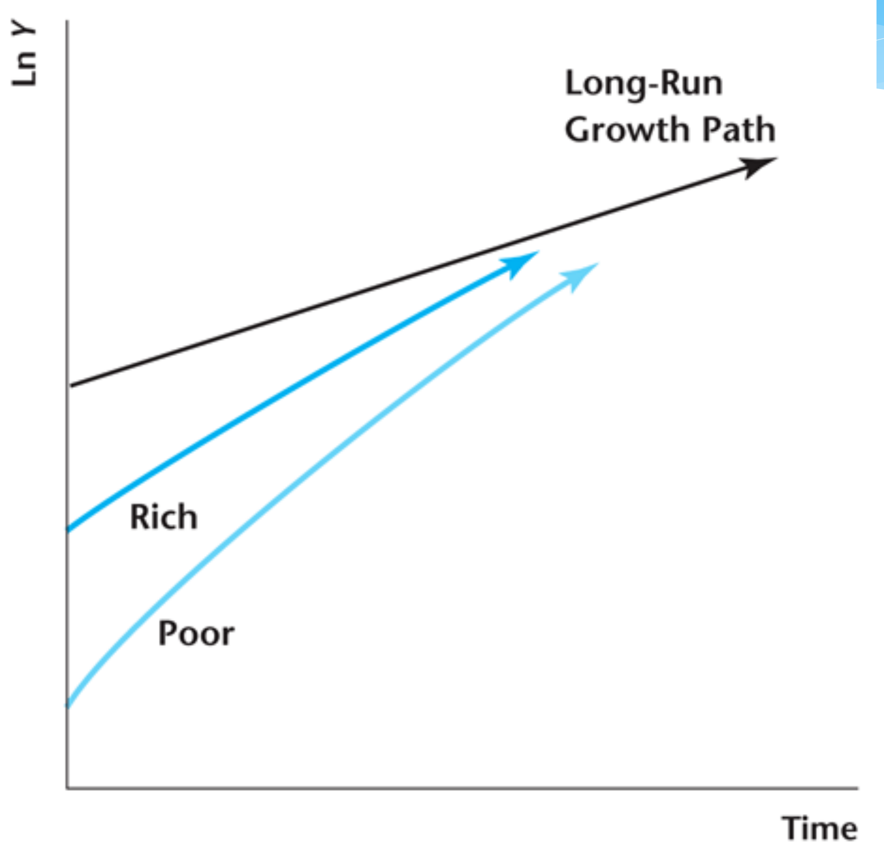


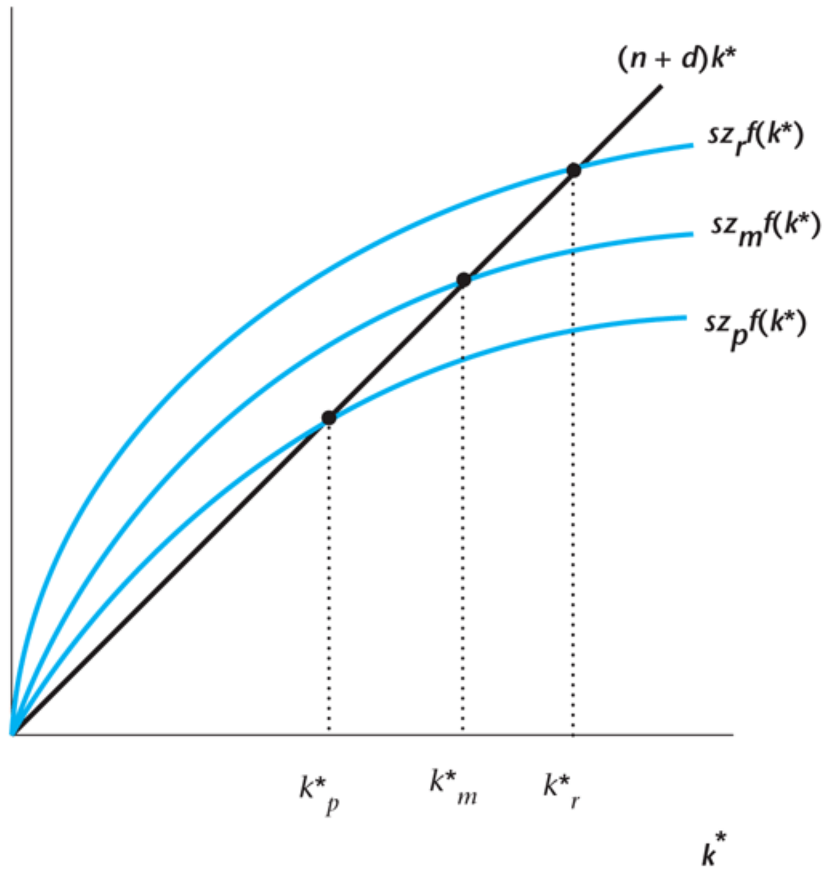
Figure 8.3
Convergence in Aggregate Output Across Countries in the Solow Growth Model



TFP may differ across countries

- There exist barriers to the adoption of new technology, and these barriers can differ across countries.
  - Labor unions
  - Barriers to entry
- Economies may differ in how efficiently factors of production are allocated across firms in the economy
  - Corruption
  - Financial imperfections

Figure 8.4
Differences in Total Factor Productivity Can Explain Disparity in Income per Worker Across Countries



Case study: Growing like China

- High output growth.
- Sustained returns on capital:
  - In spite of very high investment rates (39 percent on average), the rate of return to capital has remained stable.

Case study: Growing like China

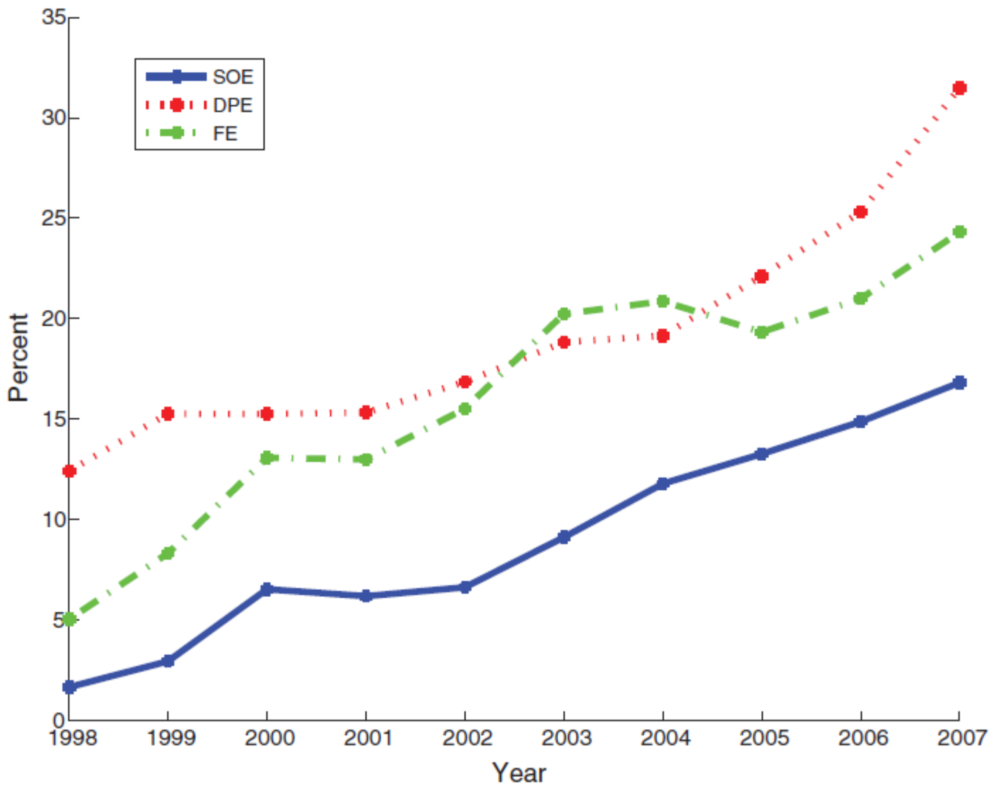


FIGURE 3. TOTAL PROFITS OVER NET VALUE OF FIXED ASSETS

Note: The figure plots the average ratio between total profits and the book value of fixed assets across firms of different ownership, in percent.

Source: CSY, various issues.

What the government can do to promote TFP Growth

- Promote competition.
- Promote free trade.
- Privatization.
- Enforcement of laws.

Endogenous Growth: A Model of Human Capital Accumulation

- A model with endogenous TFP growth.
- We are interested in the following questions:
  - How does TFP growth respond to the quantity of public funds spent on public education?
  - How is TFP growth affected by subsidies to research and development?
  - Does it make sense to have the government intervene to promote economic growth?

Human Capital

- Human capital: the accumulated stock of skills and education that a worker has at a point in time.

- Human capital
  - It is an investment;
  - The knowledge, the key input of human capital, is nonrivalry;
  - Human capital investment has constant return to scale.

Consumer Behavior

Consumption is equal to total wage income.

$$C = wuH^s$$

Human Capital Accumulation

Future human capital depends on current human capital and time devoted to training and education.

$$H^{s}{}' = b(1-u)H^{s}$$

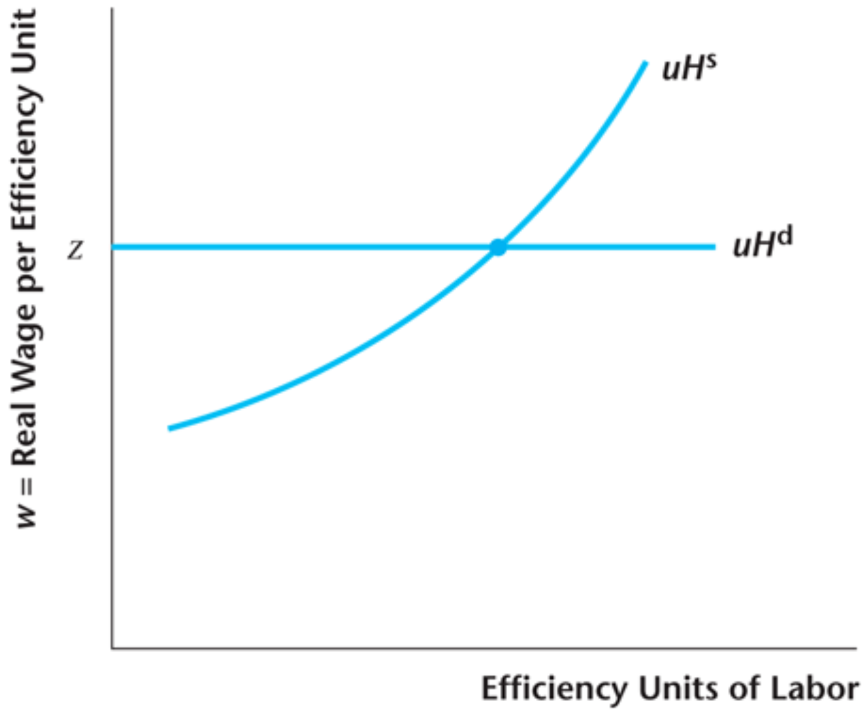
Production Function

$$Y = zuH^d,$$

Firm's Profits

$$\pi = zuH^d - wuH^d = (z - w)uH^d.$$

Figure 8.5
Determination of the Equilibrium Real Wage in the Endogenous Growth Model



Equilibrium Consumption

$$C = zuH$$

Human Capital Growth

Human capital evolves in equilibrium according to:

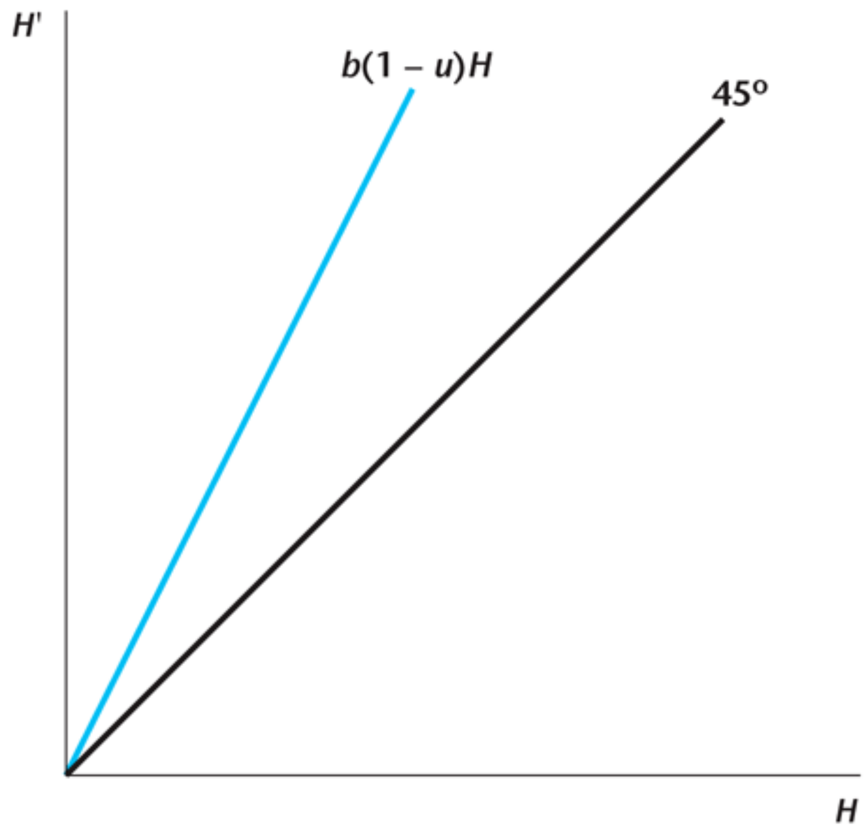
$$H' = b(1-u)H$$

Equilibrium Human Capital Growth

Equilibrium growth rate of human capital:

$$\frac{H'}{H} - 1 = b(1-u) - 1$$

Figure 8.6
Human Capital Accumulation in the Endogenous Growth Model



Economic Policy and Growth

- Change b:
  - Make the education system more efficient.
  - Provide incentive into the school system.

- Change u:
  - Provide taxes or subsidies to education

Figure 8.7
Effect of a Decrease in u on the Consumption Path in the Endogenous Growth Model

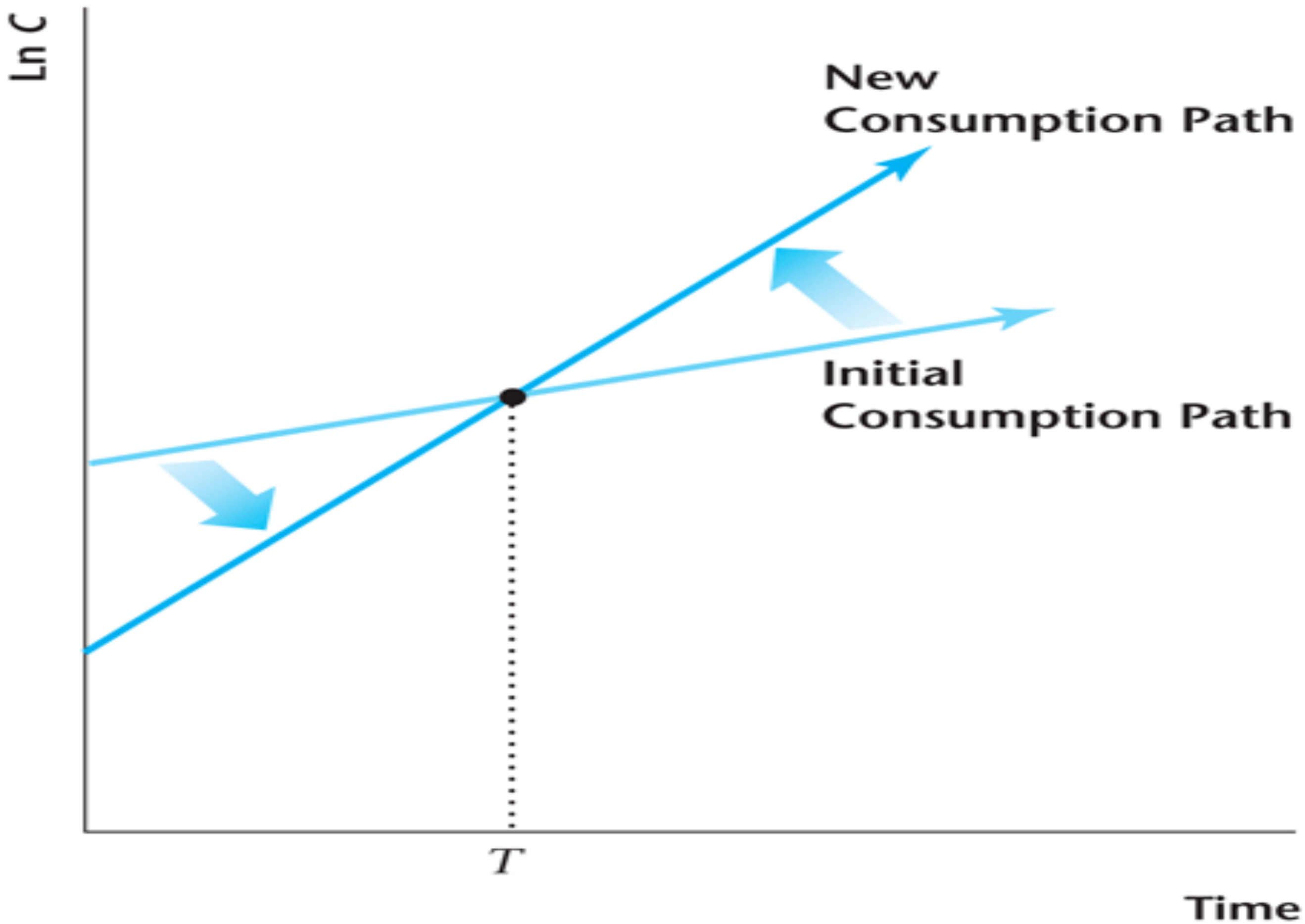
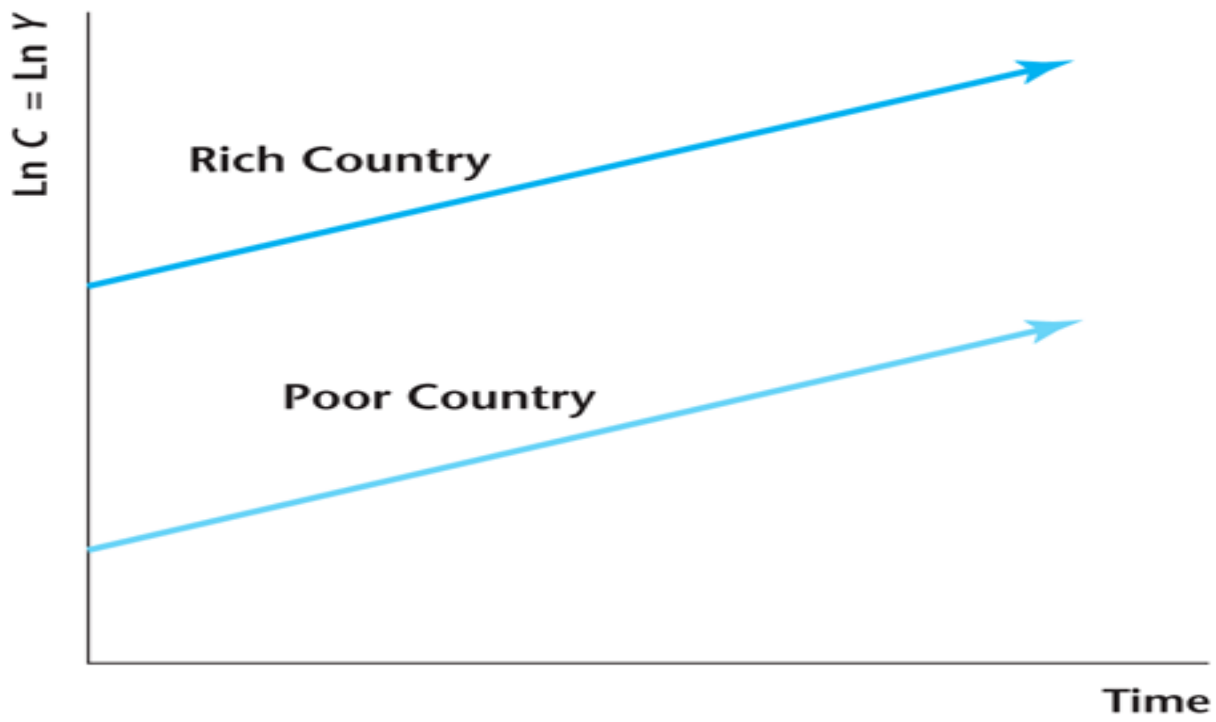


Figure 8.8
No Convergence in the Endogenous Growth Model



Practice Question

Introduce government activity in the endogenous growth model as follows. In addition to working ***u*** units of time in producing goods, the representative consumer works ***v*** units of time for the government and produces ***gvH*** goods for government use in the current period, where ***g>0***. The consumer now spends ***1-u-v*** units of time each period accumulating human capital.

a) Suppose that ***v*** increases with ***u*** decreasing by an equal amount. Determine the effects on the level and the rate of growth of consumption. Draw a diagram showing the initial path followed by the natural log of consumption and the corresponding path after v increases.

b) Suppose that v increases with u held constant. Determine the effects on the level and the rate of growth of consumption. Draw a diagram showing the initial path followed by the natural log of consumption and the corresponding path after ***v*** increases.

Reconcile with the data

- Incomes per capita among richer countries have converged.
- Human capital externalities and labor, capital mobility, eliminate the possibilities of large difference in the levels of human capital across regions .
- Brain Drain will create huge differences in human capital across very rich and very poor countries.

Reconcile with the data